WQF7003
Intelligent Computation

Course Briefing

Contact Info and Time Table

- Lecturer: Dr Lim Chee Kau
- email: limck@um.edu.my

- Classes: OCC1/2 Monday 6pm (DK1)
  OCC3 Saturday 6pm (DK1)
- Link to online classes:
  https://meet.google.com/fgc-dwir-mkb

About this course

- This is a mathematics course.
- This course does not focus on learning SOTA technologies; instead, it aims to provide a stronger mathematical foundation for fundamental research in AI, including preparation for PhD studies.
- Build on general understanding about mathematics courses at undergraduate level for Malaysia students stydying computer science, engineering and sciences.

Why we want to learn matehmatics?

Mathematics provides the theoretical foundation for modeling data, designing algorithms, and understanding how and why AI systems learn and make predictions.

Main References

- Russell, S., & Norvig, P. (2020). Artificial Intelligence: A Modern Approach, Pearson.
- Johnsonbaugh, R. (2018). Discrete mathematics, Pearson.
- Deisenroth, M. P., Faisal, A. A., & Ong, C. S. (2020). Mathematics for machine learning. Cambridge University Press.
- Johnston, N. (2021). Advanced linear and matrix algebra. Switzerland: Springer.

Mode of classes

- Physical (synchronous)
- Online (synchronous)

  https://meet.google.com/fgc-dwir-mkb
- Recorded (asynchronous, will be shared in Spectrum)

Please refer to the Spectrum before the class.

Schedule (Tentative)

Week	Date (Mon)	Date (Sat)	Topics	
1	09 Mar 26	14 Mar 26	Intro, fundamental of graphs and tre	← Online classes
2	16 Mar 26	21 Mar 26	Problem solving and uninformed sea	← Recorded videos
3	23 Mar 26	28 Mar 26	informed search and heuristic	← Recorded videos
4	30 Mar 26	04 Apr 26	game search	
5	06 Apr 26	11 Apr 26	Logic	
6	13 Apr 26	18 Apr 26	Matrix: revision	
7	20 Apr 26	25 Apr 26	Test, discuss asignment	
	27 Apr 26	02 May 26	break	
8	04 May 26	09 May 26	Vector spaces, Linear Independence	
9	11 May 26	16 May 26	Linear Transformation, Coordinate v	
10	18 May 26	23 May 26	Inner products and orthogonality	
11	25 May 26	30 May 26	Matrix decompositions	
12	01 Jun 26	06 Jun 26	Optimizations	
13	08 Jun 26	13 Jun 26	Probability	
14	15 Jun 26	20 Jun 26	Alternative assessment	

Assessments

- Final examination / Alternative assessments (50%) [15 June 2026]

- Continuous assessments 50%
  - Mid term test (30%) [20 April 2026]
  - Assignment (20%)

Communication channels

- Siswamail (official UM email) : Please make sure you check your siswamail frequently.
- Spectrum (https://spectrum.um.edu.my)
- No WhatsApp, No social media.

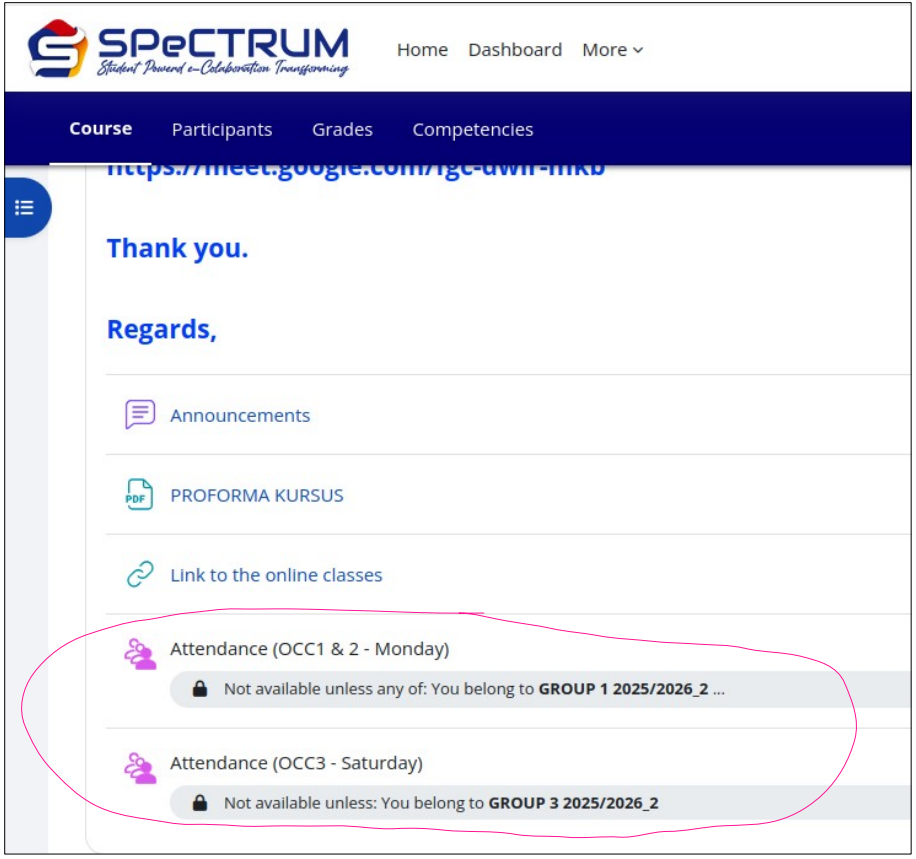
Prefer siswamail first.

Communication channels

- If you are going to contact us, please provide the following in your message:
  - Course (in this case, WIX1001)
  - OCC number
  - Name
  - Student ID
- Without the above, the reply you will receive is just a short question: “May I know who are you?” or “Which course you are from?”

Attendance

- Please log in to Spectrum to record attendance for each class.



Q&A